

GRAPHENE ; *THE MATERIAL OF THE FUTURE*

Graphene is a single layer of carbon atoms arranged in a hexagonal honeycomb structure

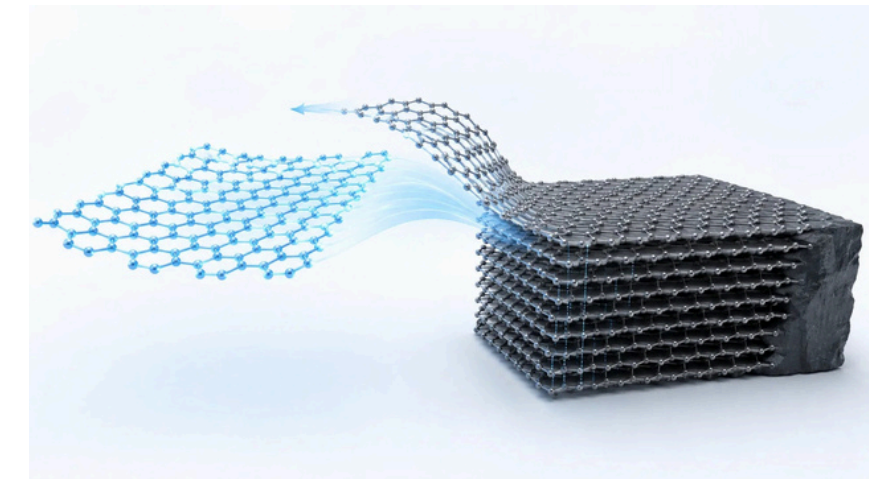
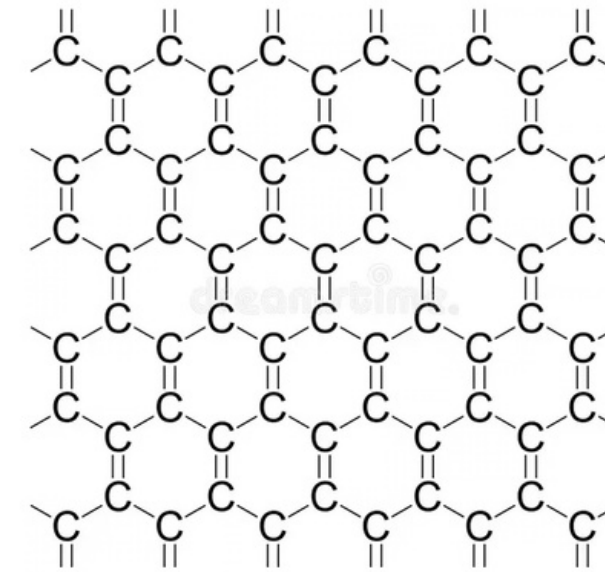
DISCOVERED IN 2004

2010 NOBEL PRIZE IN PHYSICS

WORLD'S FIRST 2D MATERIAL

HEXAGONAL CARBON LATTICE

REVOLUTIONARY WONDER MATERIAL



Celebrating 20 Years of Graphene



Andre Geim



Konstantin Novoselov

MECHANICAL & PHYSICAL PROPERTIES

The Extraordinary Characteristics of a 2D Wonder Material

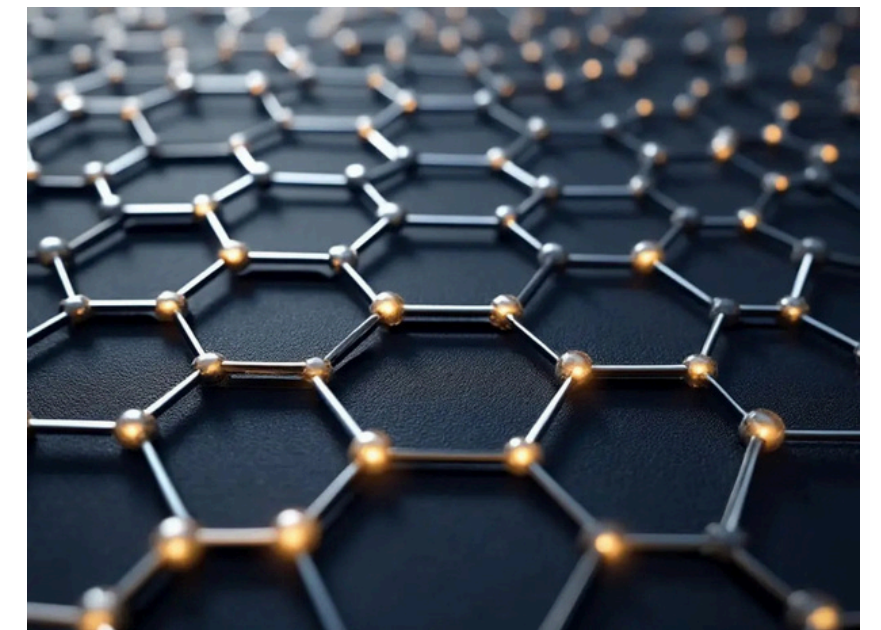
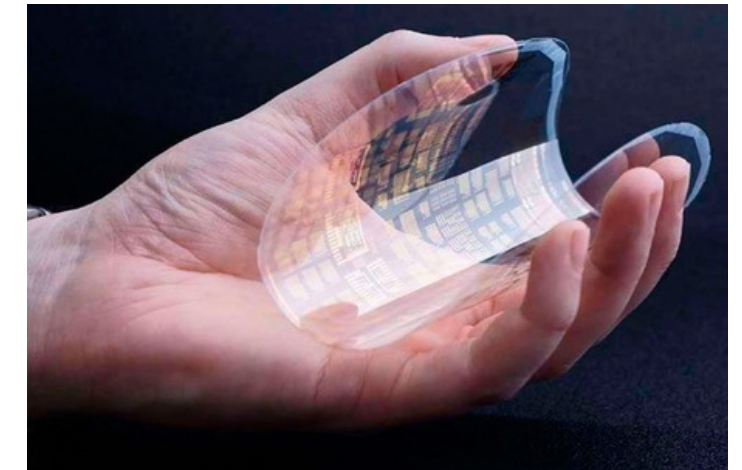
EXTREMELY STRONG

LIGHTWEIGHT

FLEXIBLE

VERY THIN

ALMOST TRANSPARENT



THERMAL & ELECTRICAL PROPERTIES

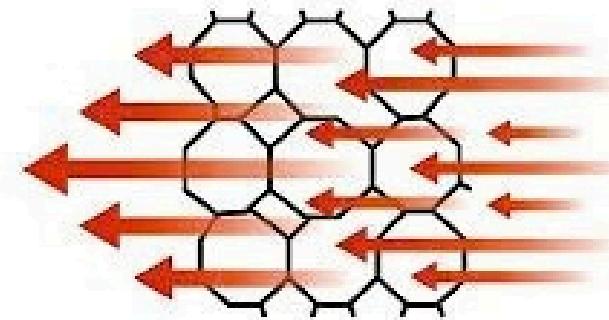
HIGH THERMAL CONDUCTIVITY

THERMAL STABILITY

HIGH ELECTRICAL CONDUCTIVITY

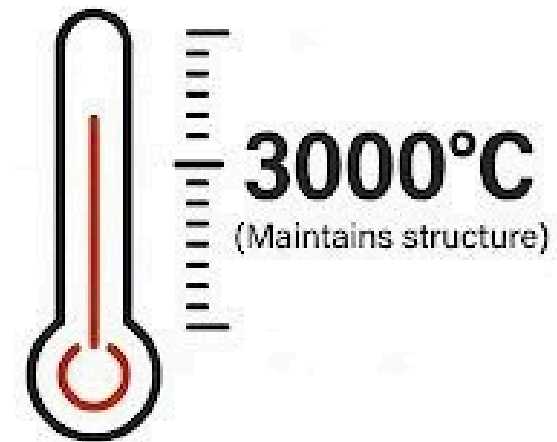
LOW RESISTIVITY

HIGH THERMAL
CONDUCTIVITY

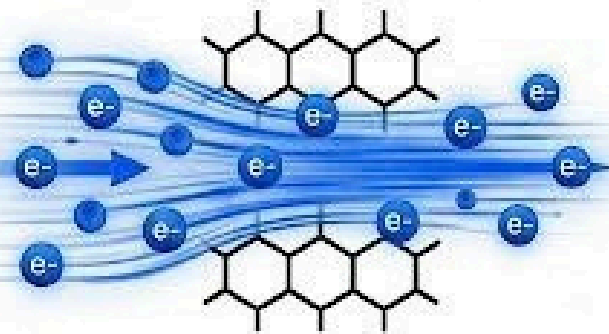


3000-5000 W/m·K
(At room temperature)

THERMAL
STABILITY

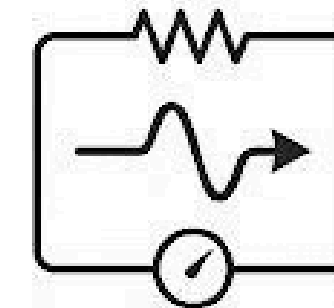


HIGH ELECTRICAL
CONDUCTIVITY



200,000 cm²/V·s
(Charge carrier mobility)

LOW RESISTIVITY

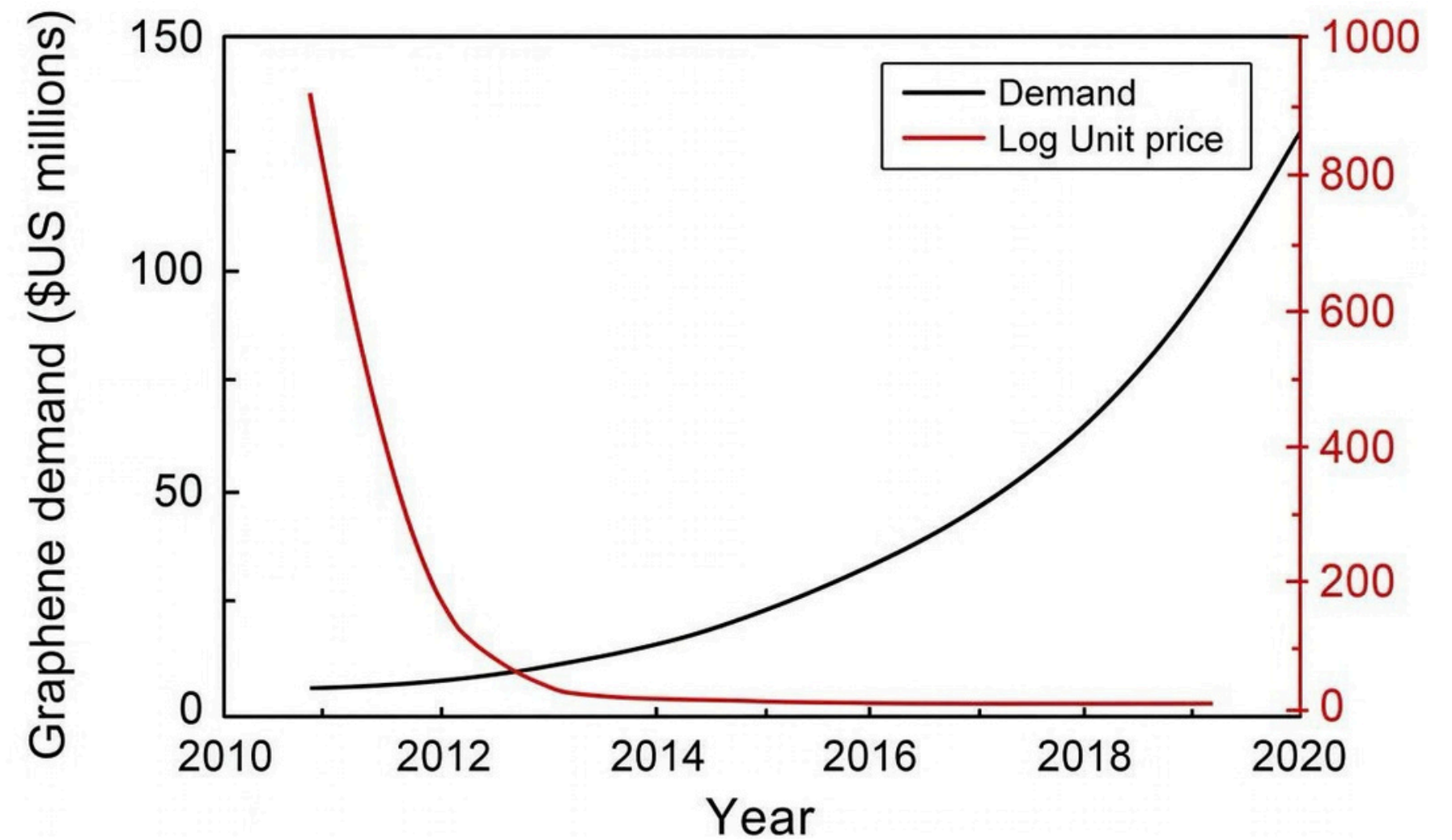
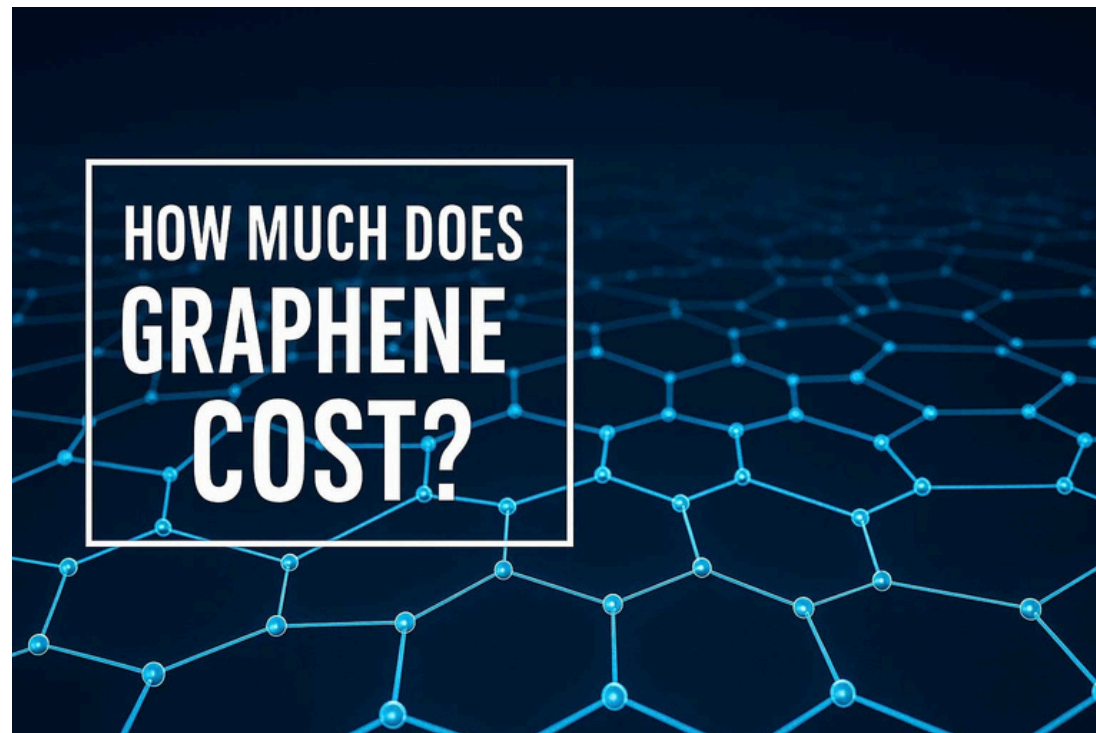


1 μΩ·cm
(Extremely efficient conductor)

COST ; VERY HIGH

HIGH PRODUCTION COST

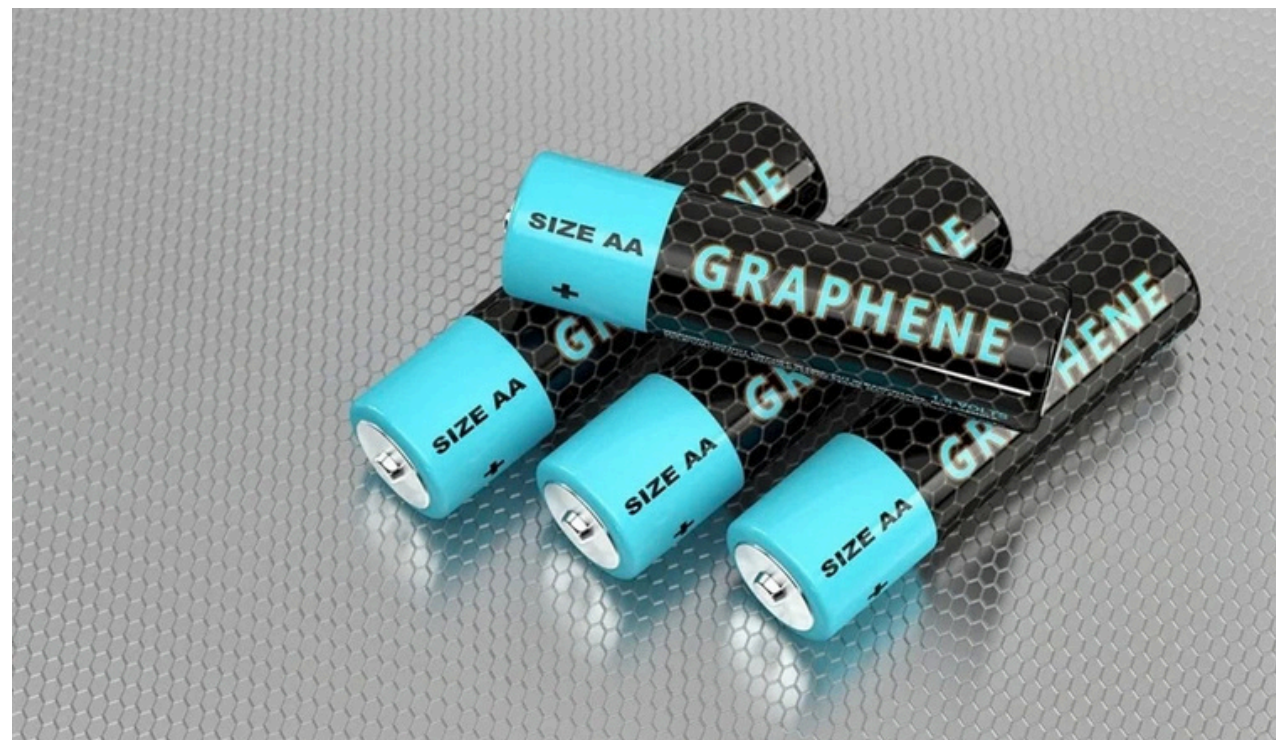
HIGH PRICE OF RAW MATERIALS



MODERN APPLICATIONS

ENERGY STORAGE AND BATTERIES

COMPOSITE MATERIALS

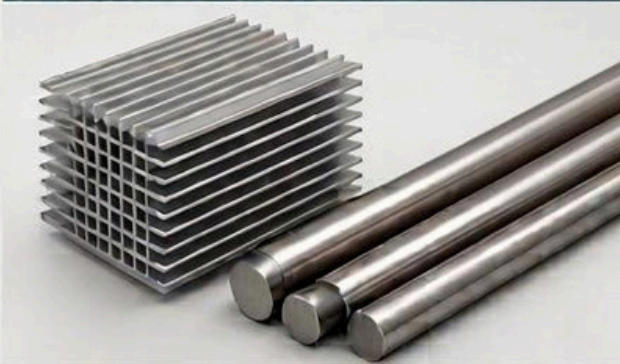


GRAPHENE-POLYMER COMPOSITES



GRAPHENE + EPOXY / PE / PP / NYLON

GRAPHENE-METAL COMPOSITES



GRAPHENE + ALUMINIUM /
COPPER / MAGNESIUM

GRAPHENE-RUBBER COMPOSITES



GRAPHENE + NATURAL / SYNTHETIC
RUBBER

GRAPHENE-CEMENT COMPOSITES



GRAPHENE + CEMENT / CONCRETE

MODERN APPLICATIONS

SENSORS

FLEXIBLE ELECTRONICS

